Exact symmetry sectors at work: finite-N lifetimes of a boundary time crystal under local dissipation and symmetry-resolved entanglement in the XXZ chain

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Abstract

When a quantum system has a symmetry, its states and its dynamics live in a block-diagonal algebra, a finite-dimensional von Neumann algebra of Type I, whose blocks are much smaller than the Hilbert space. We use this structure, as implemented in the GPU-capable PyTorch library `torch_vn_algebra`, for two exact finite-size computations. (A) For a boundary time crystal of N two-level systems with collective drive and collective decay we add local emission or local dephasing. These terms do not conserve the total spin, but they commute with the permutations of the sites. We therefore treat them as completely positive maps between the Schur–Weyl sectors $\bigoplus_j M_{2j+1} \otimes 1_{m_j}$, and evaluate the generator in $O(N^3)$ operations per state, with ten dissipation rates batched. We compute the lifetime $\tau(N, \gamma)$ of the oscillations for $N \leq 80$ without truncating the state space. Without local terms the lifetime grows linearly in N, in line with the known $1/N$ law of the damping rate. With local terms it saturates at the mean-field lifetime τ_{MF}. We observe that the collective and the local damping rates approximately add, $1/\tau \approx 1/\tau(N, 0) + 1/\tau_{\mathrm{MF}}$, with relative deviations of τ between -0.067 and $+0.172$ over all 70 data points; this is an observation, not a theorem. (B) For the ground state of the XXZ chain with $L = 24$ sites, computed for eight anisotropies in one batched Lanczos run, the $U(1)$ sector blocks of the reduced state give the symmetry-resolved entanglement directly. We determine the widths B_n of the charge distributions of ρ_A^n, test the $1/n$ law of bosonisation, measure the amplitude $G_1(\Delta)$ of the breaking of equipartition, and obtain the Luttinger parameter from the charge statistics to within 0.8 % for $-0.5 \leq \Delta \leq 0.75$. For free fermions we recover the asymptotic amplitude $1/(2\pi^2)$ of Bonsignori, Ruggiero and Calabrese and show how slowly it is approached.

Keywords: boundary time crystal; permutational invariance; Schur–Weyl duality; local dissipation; symmetry-resolved entanglement; XXZ chain; Luttinger parameter; Type I von Neumann algebra.

1 Introduction

Exact numerical results for quantum many-body systems are limited by the dimension of the Hilbert space. A symmetry reduces this dimension. If a group acts on the Hilbert space $\mathcal{H}$ and the Hamiltonian, the Lindbladian or the state commutes with it, the relevant operators lie in the commutant of the group, which in finite dimension is a direct sum of full matrix algebras with multiplicities,

$$\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{C}) \otimes 1_{m_c}, \qquad \mathcal{H} = \bigoplus_{c} \mathbb{C}^{k_c} \otimes \mathbb{C}^{m_c} \tag{1}$$

(a finite-dimensional von Neumann algebra of Type I [28]). A computation that stays in $\mathcal{M}$ only needs the blocks $A_c \in M_{k_c}$; the multiplicity spaces $\mathbb{C}^{m_c}$ never appear. Superselection rules and conserved charges lead to the same structure [2]. The idea is old. Here we take it seriously in two places where it gives data that we have not found in the literature. In both cases the decisive step is to treat an

operation that *changes* the sector, or that *splits* the system, as a map between blocks rather than as an operator on $\mathcal{H}$.

(A) Boundary time crystal with local dissipation. N two-level systems driven and damped collectively form a boundary time crystal [13]: the magnetisation oscillates, and at finite N the oscillations are damped with a rate that vanishes as $1/N$ [19, 21]. Real emitters also decay and dephase individually. Such local terms do not conserve the total spin, and in mean field they turn the limit cycle into a damped oscillation [24]. How long do the oscillations live at finite N when both mechanisms act? The local terms are permutation invariant, so the state stays in the Schur–Weyl algebra $\bigoplus_j M_{2j+1} \otimes 1_{m_j}$, of dimension $O(N^3)$ instead of 4^N. We write the local dissipators as completely positive maps between the sectors j and compute the lifetimes $\tau(N, \gamma)$ for $N \leq 80$ and ten local rates. Permutational invariance with local dissipation is exploited by several methods and codes [5, 1, 10, 15, 26]; what we add are the numbers and an observation about them: the collective and local damping rates combine approximately additively.

(B) Symmetry-resolved entanglement in the XXZ chain. For a $U(1)$-symmetric pure state the reduced state ρ_A of a block A commutes with the charge Q_A of A, so it lies in $\bigoplus_q M_{\binom{\ell}{q}}$. The blocks are the symmetry-resolved reduced states [16, 11, 31]; their entropies $S(q)$ are independent of q to leading order (equipartition [31]), and the corrections are known exactly for free fermions [3]. For the interacting XXZ chain at $L = 24$ we obtain the blocks of ρ_A directly from the sector form of the ground state, for eight anisotropies at once, and use them to measure the widths of the charge distributions, the breaking of equipartition and the Luttinger parameter.

What is known and what is new. We state the attribution explicitly, since both topics have a large literature. *Known:* the linear growth of the lifetime of the boundary time crystal with N, i.e. the $1/N$ law of its damping rate [19, 21]; the damping of the mean-field limit cycle by non-collective dissipation [24]; the permutational-invariance methods [5, 1, 10, 15, 26]; the identities that relate $S(q)$ to the charge distributions and the free-fermion asymptotics $G_1 \to 1/(2\pi^2)$ of the equipartition breaking [3]; the bosonisation background and the relation between charge fluctuations and the Luttinger parameter [11, 27, 8]. *New, as far as we know:* the finite-N lifetimes $\tau(N, \gamma)$ with local emission or dephasing for $N \leq 80$, computed exactly in the Schur–Weyl decomposition; the observed approximate additivity of the collective and local damping rates, with the structure of its deviations; the widths B_n of the charge distributions of ρ_A^n and a test of the $1/n$ law at $L = 24$; the amplitudes $G_1(\Delta)$ across anisotropies; and a determination of the Luttinger parameter from the charge statistics, to within 0.8 % for $-0.5 \leq \Delta \leq 0.75$. The computations use the library `torch_vn_algebra` [18], built on PyTorch [22]; brief accounts of both computations appear as use cases in its description [18]. All numbers below come from CPU runs of the scripts in its repository.

Organisation. Section 2 describes the block-diagonal representation, the Schur–Weyl decomposition with local dissipators as maps between sectors, and the $U(1)$ sectors of a bipartition. Section 3 treats the boundary time crystal, Section 4 the symmetry-resolved entanglement. Section 5 lists the checks, and Section 6 gives an outlook.

2 Method: computing in the blocks

2.1 Block-diagonal algebras, states and maps between sectors

Let $\mathcal{M}$ be as in (1). An element of $\mathcal{M}$ is a list of blocks $A = \bigoplus_c A_c$, and the trace of $\mathcal{H}$ restricts to $\mathrm{Tr}_{\mathcal{H}}(A \otimes 1) = \sum_c m_c \,\mathrm{Tr}\, A_c$. A state ρ on $\mathcal{H}$ that commutes with the symmetry has the form $\rho = \bigoplus_c \sigma_c \otimes 1_{m_c}/m_c$, and we store the blocks σ_c, which satisfy $\sum_c \mathrm{Tr}\,\sigma_c = 1$ and $\mathrm{Tr}_{\mathcal{H}}(\rho A) = \sum_c \mathrm{Tr}(\sigma_c A_c)$. In the language of [18], σ is the density of the state with respect to the "blunt" trace $\sum_c \mathrm{Tr}\, A_c$; $p_c = \mathrm{Tr}\,\sigma_c$ is the probability of the sector c.

Every normal completely positive map $\Phi : \mathcal{M} \to \mathcal{M}$ is a sum of maps between blocks,

$$\Phi(\sigma)_d = \sum_c \sum_i K_i^{dc}\, \sigma_c\, K_i^{dc*}, \qquad K_i^{dc} : \mathbb{C}^{k_c} \to \mathbb{C}^{k_d}. \tag{2}$$

This is the form in which a symmetric dissipator enters. A Lindblad term $\sum_i \mathcal{D}[L_i]$, $\mathcal{D}[L]\rho = L\rho L^* - \frac{1}{2}\{L^*L, \rho\}$, whose individual jump operators L_i do not commute with the symmetry but whose sum does, need not be of the form $\mathcal{D}[L]$ with $L \in \mathcal{M}$. It is, however, of the form $\Phi(\rho) - \frac{1}{2}\{\Phi^*(1), \rho\}$ with the completely positive map $\Phi(\rho) = \sum_i L_i \rho L_i^*$, which maps $\mathcal{M}$ into $\mathcal{M}$, and with $\Phi^*(1) = \sum_i L_i^* L_i \in \mathcal{M}$. The generator therefore acts on the blocks: a Hamiltonian and jump operators in $\mathcal{M}$ act inside the blocks, and a map (2) couples them. In the library this is the class `InterSectorChannel`, which `dynamics.lindblad_evolve` accepts as a jump [18].

2.2 Schur–Weyl sectors of N two-level systems

By Schur–Weyl duality [9], the symmetric group permuting the factors of $(\mathbb{C}^2)^{\otimes N}$ and the collective $SU(2)$ generate each other's commutants,

$$(\mathbb{C}^2)^{\otimes N} = \bigoplus_j V_j \otimes \mathbb{C}^{m_j}, \qquad \dim V_j = 2j+1, \quad m_j = \binom{N}{N/2-j} - \binom{N}{N/2-j-1}, \tag{3}$$

$j = N/2, N/2-1, \cdots \geq 0$. The permutation-invariant operators form $\mathcal{M} = \bigoplus_j M_{2j+1}(\mathbb{C}) \otimes 1_{m_j}$. A state of this algebra has $\sum_j (2j+1)^2 = O(N^3)$ entries, for $N = 80$ exactly 91881, whereas a density matrix on $(\mathbb{C}^2)^{\otimes 80}$ has $4^{80} \approx 1 \times 10^{48}$ entries. The blocks σ_j are the "Dicke-basis" densities of PIQS [26]. The collective operators $J_a = \frac{1}{2}\sum_i \sigma_a^{(i)}$ act in block j as the spin-j matrices in the basis $|j, m\rangle$.

Local dissipators as maps between sectors. For a single-site operator $L \in \{\sigma_-, \sigma_+, \sigma_z\}$ the map $\Phi(X) = \sum_{i=1}^N L^{(i)} X L^{(i)*}$ commutes with the permutations. To find its Kraus operators on the blocks we restrict to the last site and branch $N \to N-1$: the multiplicity space $\mathbb{C}^{m_j}$ is the direct sum of the multiplicity spaces of $j_1 = j \pm \frac{1}{2}$ for $N-1$ sites, $L^{(N)}$ preserves j_1, and all N sites contribute equally. This gives

$$\Phi(\sigma)_{j'} = \sum_j \sum_{j_1} K_{j_1}^{j'j}\, \sigma_j\, K_{j_1}^{j'j*}, \qquad K_{j_1}^{j'j} = \Big(\frac{N\, m_{j_1}^{(N-1)}}{m_j^{(N)}}\Big)^{1/2} \big\langle j_1, \tfrac{1}{2}; j' \big|\, 1 \otimes L \,\big| j_1, \tfrac{1}{2}; j \big\rangle, \tag{4}$$

with $|j' - j| \leq 1$, $j_1 \in \{j \pm \frac{1}{2}\} \cap \{j' \pm \frac{1}{2}\}$, and the coupled multiplets of $V_{j_1} \otimes \mathbb{C}^2$ (Clebsch–Gordan coefficients of $j_1 \otimes \frac{1}{2}$) in the last factor. Each K has a single non-zero diagonal, and $\Phi^*(1) = \sum_i L^{(i)*} L^{(i)}$ equals $N/2 + J_z$, $N/2 - J_z$ or N for σ_-, σ_+, σ_z. Closed forms of these coefficients are given in [5, 1, 26]; we verify ours against the exact generator on $(\mathbb{C}^2)^{\otimes N}$ and against PIQS (Section 5). The same symmetry is used by PsiQuaSP [10] and by Kirton and Keeling [15].

Banded evaluation and batching. In the library, `PermutationalSystem(N)` is the algebra with these sectors, and `local_jump` returns Φ as an `InterSectorChannel` of Kraus rank two. With it the generic integrator handles collective jumps as operators and local ones as maps between sectors. Its dense Kraus tensor has $2C^2k_{\max}^2$ entries ($C = \lfloor N/2 \rfloor + 1$ sectors, $k_{\max} = N + 1$), which limits this route to a few tens of spins. Every operator in the generators below is banded in the basis $|j, m\rangle$: $J_\pm$ and $\sigma_\pm$ shift m by one, and the local maps change j by at most one. `PermutationalLindbladian` therefore evaluates

$$\dot{\sigma} = -i[\omega_x J_x + \omega_y J_y + \omega_z J_z, \sigma] + \Gamma_- \mathcal{D}[J_-]\sigma + \Gamma_+ \mathcal{D}[J_+]\sigma + \Gamma_z \mathcal{D}[J_z]\sigma \\ + \gamma_- \sum_i \mathcal{D}[\sigma_-^{(i)}]\sigma + \gamma_+ \sum_i \mathcal{D}[\sigma_+^{(i)}]\sigma + \gamma_z \sum_i \mathcal{D}[\sigma_z^{(i)}]\sigma \quad (5)$$

as a sum of shifted elementwise products of the blocks and a tridiagonal Hamiltonian part. This costs $O(N^3)$ operations per state instead of $O(N^4)$ for dense block products, and every coefficient may be a vector, one entry per batch element. With $\omega_x = \omega_z = 0$ every term maps real matrices to real matrices, and the computation runs in the real algebra $\bigoplus_j M_{2j+1}(\mathbb{R})$. Finally, `PermutationalSystem(N, j_min)` keeps only the sectors $j \geq j_{\min}$. This compression generates a trace-decreasing completely positive semigroup whose error on an observable A is bounded by $2\|A\|$ times the lost weight. We use it only where it is exact: without local terms only the sector $j = N/2$ is populated.

2.3 $U(1)$ sectors of a bipartition

A spin-$\frac{1}{2}$ chain of L sites with a Hamiltonian that conserves S^z_{tot} is block diagonal in the number N of up spins: $\mathcal{M}_L = \bigoplus_{N=0}^{L} M_{\binom{L}{N}}(\mathbb{C})$. Its ground state at $S^z_{\text{tot}} = 0$ is a vector in the sector $N = L/2$, of dimension $\binom{L}{L/2}$ (2,704,156 for $L = 24$). For a block $A = \{0, \ldots, \ell - 1\}$ and its complement B, the sector N splits as

$$\mathbb{C}^{\binom{L}{N}} = \bigoplus_q \mathbb{C}^{\binom{\ell}{q}} \otimes \mathbb{C}^{\binom{L-\ell}{N-q}}, \quad (6)$$

the sector N of the fused tensor product $\mathcal{M}_A \otimes_f \mathcal{M}_B$ of the two chain algebras, in which only the total charge is conserved (`composite.fused_tensor_product` in [18]). In this layout $|\psi\rangle$ is a list of matrices $\Psi_q \in \mathbb{C}^{\binom{\ell}{q} \times \binom{L-\ell}{N-q}}$, and

$$\rho_A = \operatorname{Tr}_B |\psi\rangle\langle\psi| = \bigoplus_{q=0}^{\ell} \Psi_q \Psi_q^* \ \in \ \mathcal{M}_A = \bigoplus_q M_{\binom{\ell}{q}}(\mathbb{C}). \quad (7)$$

The matrix $|\psi\rangle\langle\psi|$, of dimension $\binom{24}{12}^2 \approx 7.3 \times 10^{12}$ at $L = 24$, is never formed. The minimal central projections of $\mathcal{M}_A$ are the spectral projections Π_q of Q_A, so the sector probabilities $p(q) = \operatorname{Tr} \Psi_q \Psi_q^*$ and the spectra of the normalised blocks are exactly the symmetry-resolved data.

For the XXZ chain the anisotropy Δ multiplies only the diagonal $S^z S^z$ part of the sector Hamiltonian. The Hamiltonians for several Δ therefore form one batched sparse operator with a shared hopping part (`krylov.SparseSectorHamiltonian`), and one batched Lanczos run gives all ground states.

3 Part A: a boundary time crystal under local dissipation

3.1 Model and observables

We consider

$$\dot\rho = -i[\omega J_x, \rho] + \frac{\kappa}{S}\,\mathcal{D}[J_-]\rho + \gamma \sum_{i=1}^{N} \mathcal{D}[\sigma_-^{(i)}]\rho + \gamma_\phi \sum_{i=1}^{N} \mathcal{D}[\sigma_z^{(i)}]\rho, \qquad S = \frac{N}{2}. \tag{8}$$

For $\gamma = \gamma_\phi = 0$ this is the boundary time crystal of Iemini et al. [13]. For $\omega > \kappa$ the magnetisation $m_z = \langle J_z\rangle/S$ oscillates, and at finite N the oscillations are damped with a rate that vanishes as $1/N$; for $\omega \gg \kappa$ the rate is $3\kappa/(2N)$ for oscillations from all spins down [19, 21]. The mean-field equations become exact as $N \to \infty$ at fixed times; Carollo and Lesanovsky [4] prove this for open Dicke models and note that the proof adapts to all-to-all spin models and to individual spin dissipation. In mean field, non-collective dissipation damps the limit cycle [24]. Robustness of boundary time crystals has also been studied in multi-level settings [25], and single-particle dephasing in a related synchronised phase [29].

We take $\omega = 2\kappa$ and all spins down at $t = 0$. In the computation we drive about y instead of x. The rotation $e^{-i\pi J_z/2}$ maps one drive to the other and commutes with every dissipator in (8), so $m_z(t)$ is unchanged, and the generator becomes real (Section 2.2).

Lifetime. Let $t_0 = 0 < t_1 < t_2 < \dots$ be the extrema of $m_z(t)$ and $a_k = |m_z(t_{k+1}) - m_z(t_k)|/2$ the half-swings. The lifetime τ is the first time at which a_k falls below a_0/e, with log-linear interpolation between the midpoints $(t_k + t_{k+1})/2$ of consecutive swings. A run that ends before this happens is censored.

Mean-field limit. For $N \to \infty$, $m = \langle \mathbf{J}\rangle/S$ obeys (drive about y)

$$\dot m_x = \omega m_z + \kappa m_x m_z - \Gamma_2 m_x, \quad \dot m_y = \kappa m_y m_z - \Gamma_2 m_y, \quad \dot m_z = -\omega m_x - \kappa(m_x^2 + m_y^2) - \gamma(1 + m_z), \tag{9}$$

with $\Gamma_2 = \gamma/2 + 2\gamma_\phi$. Without local terms the orbit from $m = (0, 0, -1)$ is closed and m_z does not decay. We define $\tau_{\mathrm{MF}}(\gamma)$ and $\tau_{\mathrm{MF}}(\gamma_\phi)$ as the lifetimes of the solutions of (9), integrated to $\kappa t = 600$.

Protocol. For every $N \in \{10, 20, 30, 40, 50, 60, 80\}$ one batched run holds the 10 parameter sets $\gamma/\kappa \in \{0.02, 0.05, 0.1, 0.2, 0.5\}$ (with $\gamma_\phi = 0$) and $\gamma_\phi/\kappa \in \{0.005, 0.0125, 0.025, 0.05, 0.125\}$ (with $\gamma = 0$). The two lists are paired so that $\gamma = 4\gamma_\phi$, i.e. equal transverse rates Γ_2. The generator is integrated with the classical fourth-order Runge–Kutta method (RK4) in the real algebra $\bigoplus_j M_{2j+1}(\mathbb{R})$, with all $N/2+1$ (or $(N+1)/2$) sectors kept. The step is $h \le 3.5/r$, with r an upper bound on the spectral radius of the generator, and m_z is sampled every $\kappa\,\delta t = 0.05$. A batch element is dropped once its lifetime is settled ($t > \tau + 6/\kappa$), and the run ends at $\kappa t = 45$. The runs without local terms use the single block M_{N+1} ($j_{\min} = N/2$) and last until $\kappa t = 0.6N + 15$.

3.2 Results

Table 1 and Figure 1 summarise the sweep; the largest run, $N = 80$ with 10 parameter sets, took 690 s on one CPU thread.

Table 1: Lifetime $\kappa\tau$ of the oscillations of m_z in (8), $\omega = 2\kappa$, from all spins down: without local terms (column 0), with local emission γ/κ and with local dephasing γ_ϕ/κ. Columns at equal position in the two groups have equal $\Gamma_2 = \gamma/2 + 2\gamma_\phi$. Last row: mean field ($N \to \infty$), Eq. (9). No entry is censored.

		local emission γ/κ					local dephasing γ_ϕ/κ				
N	0	0.02	0.05	0.1	0.2	0.5	0.005	0.0125	0.025	0.05	0.125
10	5.49	5.19	4.76	4.20	3.46	2.62	5.39	5.26	5.10	4.73	3.88
20	9.98	8.90	7.71	6.30	4.66	3.00	9.60	9.12	8.46	7.37	5.33
30	14.52	12.18	10.04	7.76	5.38	3.19	13.54	12.51	11.19	9.31	6.12
40	19.06	15.05	11.88	8.79	5.86	3.31	17.16	15.46	13.46	10.74	6.65
50	23.57	17.52	13.36	9.55	6.17	3.38	20.51	18.07	15.37	11.84	6.99
60	28.07	19.73	14.61	10.15	6.40	3.45	23.58	20.39	17.01	12.72	7.27
80	37.08	23.51	16.54	10.99	6.72	3.52	29.08	24.41	19.68	14.05	7.63
∞ (MF)	–	68.35	28.03	14.56	7.84	3.77	195.67	78.96	40.13	20.64	8.88

Without local terms the lifetime grows linearly. A least-squares fit for $20 \le N \le 80$ gives

$$\kappa\tau(N, 0) \approx 0.452\, N + 0.98. \tag{10}$$

The linear growth is the known $1/N$ law of the damping rate [19, 21]. In the perturbative regime $\omega \gg \kappa$ the rate $3\kappa/(2N)$ of [19, 21] corresponds to $\kappa\tau \approx 2N/3$. Our $\omega = 2\kappa$ lies outside that regime, and the slope 0.452 is smaller. Only this value of the slope at $\omega = 2\kappa$ is new.

With local terms the lifetime saturates at the mean-field value as N grows (last row of Table 1). For example, with $\gamma = 0.5\kappa$ the lifetime grows only from 2.62 to 3.52 between $N = 10$ and $N = 80$, against $\kappa\tau_{\rm MF} = 3.77$. Local emission damps more strongly than local dephasing at equal transverse rate Γ_2 (compare $\gamma = 0.1\kappa$, $\kappa\tau_{\rm MF} = 14.56$, with $\gamma_\phi = 0.025\kappa$, $\kappa\tau_{\rm MF} = 40.13$). In (9), emission also relaxes m_z directly through the term $-\gamma(1 + m_z)$.

3.3 Approximate additivity of the damping rates

The collective damping vanishes as $N \to \infty$; the local damping does not. We find that the two combine approximately like independent decay rates,

$$\frac{1}{\tau(N, \gamma)} \approx \frac{1}{\tau_{\rm rule}} = \frac{1}{\tau(N, 0)} + \frac{1}{\tau_{\rm MF}(\gamma)}, \tag{11}$$

for local emission and local dephasing alike. In the variables $x = \tau(N, 0)/\tau_{\rm MF}$ and $\tau/\tau_{\rm MF}$, (11) is the single curve $x/(1 + x)$. All data lie close to it (Figure 1c), although x spans more than two decades. The crossover at which the local damping takes over, $x = 1$, lies at $N^* \approx \kappa\tau_{\rm MF}/0.452$ by (10). For instance, for $\gamma = 0.1\kappa$ we have $\kappa\tau(30, 0) = 14.52$ against $\kappa\tau_{\rm MF} = 14.56$. The rule then predicts $\kappa\tau = 7.27$ at $N = 30$, and we find 7.76.

Size and structure of the deviations. Over the 70 entries of Table 1 with finite τ and $\tau_{\rm MF}$, $\tau/\tau_{\rm rule} - 1$ lies between -0.067 and $+0.172$, and between -0.067 and $+0.110$ for $N \ge 30$. The deviation has a structure (Figure 1d):

- for the larger rates τ exceeds $\tau_{\rm rule}$, by an amount that decreases at large N (for $\gamma = 0.5\kappa$ from $+17\,\%$ at $N = 10$ to $+3\,\%$ at $N = 80$);

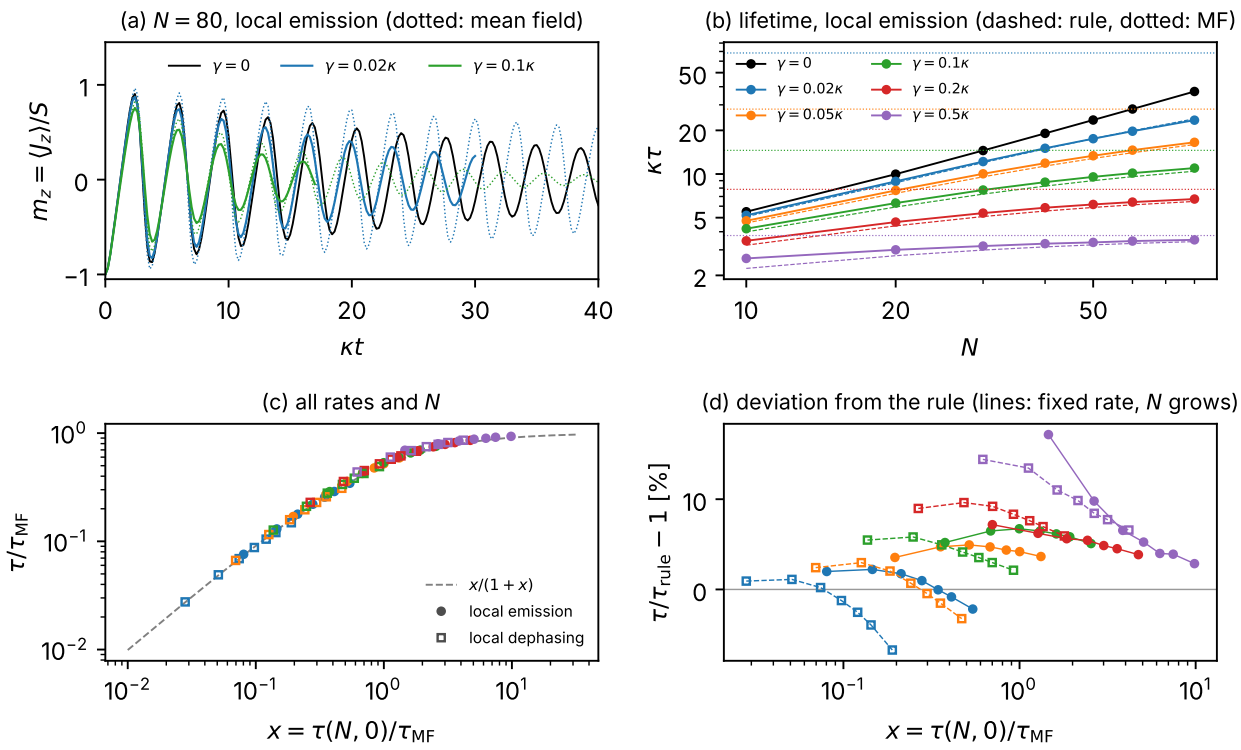


Figure 1: (a) $m_z(t)$ for $N = 80$ without and with local emission ($\gamma = 0.02\kappa$, 0.1κ), and the mean-field curves (dotted). (b) $\tau(N, \gamma)$ for local emission; dotted: $\tau_{\mathrm{MF}}(\gamma)$, dashed: the additive rule (11). (c) All lifetimes with a finite mean-field limit, τ/τ_{MF} against $x = \tau(N, 0)/\tau_{\mathrm{MF}}$; the dashed curve $x/(1 + x)$ is (11). (d) The relative deviation from the rule; points of equal rate are joined, N increasing to the right (filled: emission, open: dephasing).

- for the smallest rates the deviation decreases through zero as N grows, so that $\tau < \tau_{\mathrm{rule}}$ at the largest N (for $\gamma_\phi = 0.005\kappa$ it reaches -0.067 at $N = 80$);
- at equal x, local dephasing deviates more than local emission.

A heuristic reading of (11) is that the envelope of the oscillation is the product of a collective and a local decay factor; if both were exponential, their rates would add, up to the offset by the midpoint of the first swing that our definition of τ contains. The deviations indicate that the decay is not simply a product of two independent exponentials. We therefore state (11) as an *observation* for this model, these rates, $\omega = 2\kappa$ and this definition of the lifetime, valid to within the range just given. We have no proof of it, and our data do not establish it as an asymptotic law for $N \to \infty$ at fixed rate. The trend of the smallest rates suggests that the deviations at larger N need not stay inside the range found here.

3.4 Relation to known results

Without local terms, the growth of the lifetime with N is the known $1/N$ law of the damping rate [19, 21]; our contribution here is only the slope (10) at $\omega = 2\kappa$. The damping of the mean-field limit cycle by non-collective dissipation is due to Piccitto et al. [24]; for local terms it follows directly from (9), which is how we compute τ_{MF}. That local dephasing and local losses wash out the oscillations at finite N was shown with PIQS for $N = 30$ [26]. New, as far as we know, are

the finite-N lifetimes with local emission or dephasing for N up to 80 in Table 1, the approximate additivity (11) and the structure of its deviations. The superradiance checks of Section 5 reproduce textbook results [7, 12].

4 Part B: symmetry-resolved entanglement in the XXZ chain

4.1 Model and resolved quantities

We consider

$$H = \sum_{\langle ij \rangle} \big(S_i^x S_j^x + S_i^y S_j^y + \Delta S_i^z S_j^z \big), \qquad -1 < \Delta \le 1, \tag{12}$$

on $L = 24$ sites with periodic or open boundary conditions, and its ground state in the sector $S^z_{\mathrm{tot}} = 0$. The low-energy theory is a Luttinger liquid with the Bethe-ansatz parameter $K = \pi/(2(\pi - \arccos\Delta))$, so $K = 1$ for free fermions ($\Delta = 0$) and $K = \frac{1}{2}$ at the isotropic point. We use $\Delta \in \{-0.75, -0.5, -0.25, 0, 0.25, 0.5, 0.75, 1\}$.

For $A = \{0, \dots, \ell - 1\}$, $\rho_A = \bigoplus_q p(q) \rho_A^{(q)}$ by (7), and with $S(q) = S(\rho_A^{(q)})$ the von Neumann entropy splits as

$$S(\rho_A) = \sum_q p(q) S(q) + H(p), \tag{13}$$

into the configurational entanglement and the *number entanglement* $H(p)$, the Shannon entropy of the charge distribution [16, 11, 31]. The number entanglement has been measured with cold atoms [20] and used as a probe of many-body localisation [14]. Goldstein and Sela [11] obtained the resolved entropies from the charged moments $Z_n(\alpha) = \operatorname{Tr} \rho_A^n e^{i\alpha Q_A}$. Xavier, Alcaraz and Sierra [31] found equipartition: $S(q)$ is independent of q to leading order in ℓ. For Luttinger liquids the corrections decay logarithmically with non-universal prefactors [8], and the variance of Q_A (the full counting statistics) grows as $(K/\pi^2) \log \ell$ [27].

Identities. Let $Z_n(q) = \operatorname{Tr} \Pi_q \rho_A^n$ and $p_n(q) = Z_n(q) / \sum_{q'} Z_n(q')$, the distribution of Q_A in $\rho_A^n / \operatorname{Tr} \rho_A^n$, so $p_1 = p$. With $S_n(q) = S_n(\rho_A^{(q)})$, the expressions $S_n(q) = (1-n)^{-1} \log[Z_n(q)/Z_1(q)^n]$ and $S(q) = -\partial_n [Z_n(q)/Z_1(q)^n]_{n=1}$ of Bonsignori, Ruggiero and Calabrese [3, Eq. (2.6)] read, in terms of p_n,

$$S_n(q) - S_n(\rho_A) = \frac{\log p_n(q) - n \log p(q)}{1 - n}, \qquad S(q) - S(\rho_A) = \log p(q) - \frac{\partial}{\partial n} \log p_n(q) \Big|_{n=1}. \tag{14}$$

The q-dependence of the resolved entropies is thus the n-dependence of the charge distributions.

Gaussian mechanism and the widths B_n. At half filling $\langle Q_A \rangle = \ell/2$. Put $q_0 = \lceil \ell/2 \rceil$, $x = q - \ell/2$ and $x_0 = q_0 - \ell/2 \in \{0, \frac{1}{2}\}$. If $p_n(q) \propto e^{-x^2/2B_n}$ on the integers, (14) gives

$$S_n(q) - S_n(q_0) = -\frac{G_n \, (x^2 - x_0^2)}{2 B_1 B_n}, \qquad G_n = \frac{n B_n - B_1}{n - 1}, \qquad G_1 = \frac{\partial}{\partial n} (n B_n) \Big|_{n=1}. \tag{15}$$

Bosonisation gives this form: $Z_n(\alpha)/Z_n(0) \simeq e^{i\alpha\ell/2 - \alpha^2 B_n/2}$ for $|\alpha| < \pi$, whose Fourier series is the integer-sampled Gaussian, with

$$B_n = \frac{K}{\pi^2 n} \log \ell_c + \gamma_n + o(1), \qquad \ell_c = \frac{L}{\pi} \sin \frac{\pi \ell}{L} \ \text{(periodic)}, \tag{16}$$

Table 2: Luttinger parameter from the charge statistics of the ground state, $L = 24$, blocks $\ell \le 12$. Column 3: fit of $b_1(\ell) = (K/\pi^2)\log \ell_c + \gamma + C(-1)^\ell \ell_c^{-2K}$ (periodic, $\ell \ge 3$, oscillating term as in [27], exponent fixed self-consistently). Columns 4–6: fits of $nB_n(\ell) = (K/\pi^2)\log \ell_c + \gamma + C(-1)^\ell$, $n = 1, 2, 3$ (periodic, $\ell \ge 4$). Column 7: the same for B_1 in the open chain, with slope $K/(2\pi^2)$.

Δ	K (Bethe)	b_1	B_1	$2B_2$	$3B_3$	B_1, open
-0.75	2.173	2.267	2.150	2.172	2.175	2.032
-0.50	1.500	1.512	1.473	1.500	1.505	1.437
-0.25	1.192	1.189	1.178	1.210	1.217	1.167
$+0.00$	1.000	0.994	0.999	1.036	1.045	0.992
$+0.25$	0.861	0.854	0.871	0.914	0.926	0.854
$+0.50$	0.750	0.744	0.769	0.819	0.833	0.737
$+0.75$	0.649	0.649	0.674	0.731	0.748	0.636
$+1.00$	0.500	0.557	0.572	0.630	0.652	0.547

and non-universal constants γ_n [11, 8]. For open chains with A at the boundary the slope is $K/(2\pi^2)$ and $\ell_c = \frac{2L}{\pi}\sin\frac{\pi\ell}{L}$. Hence nB_n has the slope K/π^2 for every n (the $1/n$ *law*), $G_n = O(1)$, and (15) is equipartition broken at order $x^2/\log^2 \ell_c$ with the non-universal amplitude $\pi^4 G_1/(2K^2)$.

We *define* B_n through the two central sectors,

$$B_n = \frac{2x_0 + 1}{2\log[p_n(q_0)/p_n(q_0+1)]}. \tag{17}$$

Then (15) holds at $q_0 \pm 1$ by construction, and by (14) $G_1 = B_1[S(q_0) - S(q_0+1)]/\log[p(q_0)/p(q_0+1)]$. The Gaussian form is tested at $q_0 \pm 2$, where for even ℓ it predicts the ratio $[S(q_0) - S(q_0+2)]/[S(q_0) - S(q_0+1)] = 4$. We also report the variance b_1 of p (the full counting statistics). It differs from B_1 because Q_A is integer valued, and the variances b_n of p_n for $n \ge 2$ are dominated by this discreteness at our sizes.

Computation. For each boundary condition the eight ground states come from batched Lanczos runs (batches of 4, Krylov dimension 40, residuals $\le 3 \times 10^{-12}$) in the sector of dimension 2,704,156. For every block $\ell = 1, \ldots, 12$ the reduced state (7) is formed from the pair blocks of the fused product, and $p(q)$, $S(q)$, $Z_n(q)$ $(n = 2, 3)$, $H(p)$, b_n, B_n and G_n follow from the spectra of the blocks, for the whole batch at once. The ground states took 908 s in total, the statistics of all blocks 45 s, and the whole run with validation 1327 s (CPU (2 threads)). Without the $U(1)$ structure the Lanczos vectors would be 6.2 times longer, a dense Hamiltonian would take 2.3 PB (58 TB even in the sector), and diagonalising the dense $2^\ell \times 2^\ell$ matrix ρ_A at $\ell = 12$ would cost 34 times more than its blocks, which moreover give $p(q)$ and $S(q)$ directly.

4.2 Results

Tables 2 and 3 and Figure 2 contain the results.

Number entanglement. At these sizes most of the entanglement is number entanglement. $H(p)/S$ lies between 0.58 and 0.87 and increases with K, and $H(p)$ equals the entropy H_{G} of the integer-sampled Gaussian of width B_1 to within 0.015 (Table 3). The total entropy S at $\ell = L/2$ varies little with Δ; the split (13) moves weight from $H(p)$ to the configurational part as K decreases.

Table 3: Periodic chain, $L = 24$, $\ell = 12$: entanglement entropy S, number entanglement $H(p)$, entropy H_{G} of the integer-sampled Gaussian of width B_1; spread $= \max_n nB_n - \min_n nB_n$ $(n = 1, 2, 3)$ in per cent of B_1; G_1 at $\ell = 6$ and $\ell = 12$; $\delta S = S(q_0) - S(q_0 \pm 1)$; ratio $= [S(q_0) - S(q_0 \pm 2)]/\delta S$ (4 if p_n is Gaussian); δS_2: δS for the Rényi-2 entropies $S_2(q)$.

Δ	S	$H(p)$	H_{G}	B_1	spread	$G_1(6)$	$G_1(12)$	δS	ratio	δS_2
-0.75	1.439	1.247	1.259	0.727	0.8	0.0184	0.0195	0.0184	3.60	0.0107
-0.50	1.415	1.132	1.146	0.580	1.6	0.0262	0.0273	0.0406	3.28	0.0261
-0.25	1.406	1.059	1.075	0.503	2.5	0.0326	0.0337	0.0666	2.96	0.0456
$+0.00$	1.405	1.004	1.019	0.452	3.6	0.0383	0.0395	0.0969	2.65	0.0699
$+0.25$	1.406	0.957	0.971	0.412	4.8	0.0434	0.0447	0.1317	2.37	0.0998
$+0.50$	1.409	0.913	0.927	0.379	6.1	0.0474	0.0491	0.1708	2.13	0.1364
$+0.75$	1.413	0.869	0.880	0.349	7.7	0.0498	0.0518	0.2128	1.94	0.1801
$+1.00$	1.415	0.820	0.828	0.319	9.4	0.0494	0.0512	0.2520	1.82	0.2300

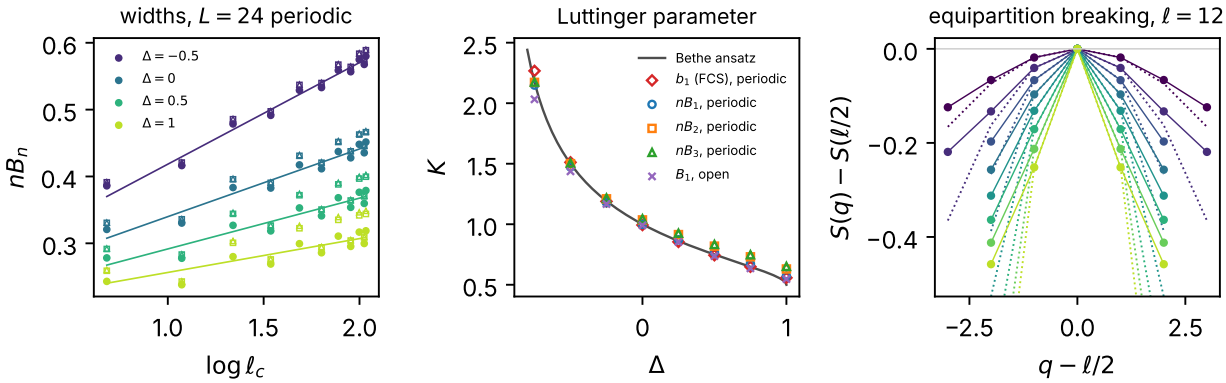


Figure 2: Periodic chain, $L = 24$. Left: nB_n against $\log \ell_c$ for $n = 1$ (filled circles), 2 (squares) and 3 (triangles); lines: slope K/π^2 (Bethe ansatz). Middle: K from the fits of Table 2. Right: $S(q) - S(\ell/2)$ at $\ell = 12$ for $\Delta \in \{-0.75, -0.5, -0.25, 0, 0.25, 0.5, 0.75, 1\}$ (dark to light); dotted: the Gaussian form (15), exact at $q - \ell/2 = \pm 1$.

Widths and the $1/n$ law. The widths B_n of (17) grow linearly in $\log \ell_c$ (Figure 2, left). The $1/n$ law holds well for large K and degrades as K decreases. The spread of nB_n over $n = 1, 2, 3$ grows from 0.8 % of B_1 at $\Delta = -0.75$ to 9.4 % at $\Delta = 1$ (Table 3). The slopes of $2B_2$ and $3B_3$ give K to within 0.1 % at $\Delta = -0.75$ (2.172 and 2.175 against 2.173), but exceed K/π^2 by 9 % and 11 % at $\Delta = 0.5$, and give 0.652 for $n = 3$ at $\Delta = 1$ (Table 2). We attribute this to finite-size corrections, which our fits do not model. In contrast, the variances b_n for $n \geq 2$ are dominated by the discreteness of Q_A; the widths B_n are the quantities in which the $1/n$ law is visible at $L \leq 24$.

Breaking of equipartition. Equipartition is clearly broken at $L = 24$. $\delta S = S(q_0) - S(q_0 \pm 1) = G_1/(2B_1^2)$ ranges from 0.018 to 0.252 and increases as K decreases (Figure 2, right). The Rényi-2 analogue δS_2 behaves in the same way. In agreement with the mechanism of Section 4.1, G_1 is nearly independent of ℓ while B_1 grows: over the even blocks $4 \leq \ell \leq L/2$ the slope of G_1 against $\log \ell_c$ is at most 10 % of that of B_1, and G_1 changes by at most 6 % between $\ell = 6$ and $\ell = 12$. The finite-size amplitude $G_1(\Delta)$ at $\ell = 12$ increases from 0.0195 at $\Delta = -0.75$ to 0.0518 at $\Delta = 0.75$, with 0.0512 at $\Delta = 1$ (Table 3). The quadratic dependence on q is not yet reached: the ratio in Table 3 is 1.82–3.60

instead of 4, and it decreases as K decreases.

Luttinger parameter. The full counting statistics b_1 give K to within 0.8 % for $-0.5 \le \Delta \le 0.75$. At $\Delta = -0.75$ the fit gives 2.267 against 2.173 (about 4 %), and at the isotropic point 0.557 instead of 0.5. There the marginal operator causes logarithmic corrections, as in the analysis of [27], which we reproduce here by exact diagonalisation. The widths B_1 give K to within 4 % for $\Delta \le 0.75$ in the periodic chain. The open chain, with a single entangling point, deviates more at negative Δ (Table 2).

4.3 Free fermions and comparison with Bonsignori, Ruggiero and Calabrese

At $\Delta = 0$ the chain is a free-fermion model, and the resolved quantities can be computed for much larger blocks from the correlation matrix (Section 5). Bonsignori, Ruggiero and Calabrese [3] obtained the Fisher–Hartwig asymptotics of the charged moments. Their Gaussian (Eq. (4.5) of [3]), written in the variables of (15), has $G_1 \to 1/(2\pi^2)$, and the breaking of equipartition at half filling is [3, Eq. (4.13)]

$$S(q) - S(\langle Q_A \rangle) \simeq -\frac{\pi^2}{4}\,\frac{(q - \langle Q_A \rangle)^2}{(\log 2\ell + 1 + \gamma_E)^2}. \tag{18}$$

These are known results. We use them as a reference and measure the approach to them.

- *Infinite XX chain.* For $\ell = 3200$, $b_1 - (\log 2\ell + 1 + \gamma_E)/\pi^2 = 2 \times 10^{-9}$, and $G_1^{(b)} = \partial_n (n b_n)|_{n=1} = 0.0506605$, while $1/(2\pi^2) = 0.0506606$. (Here $G_1^{(b)}$ is defined with the variances b_n, which the script evaluates in closed form over the entanglement spectrum.) The exact $S(q_0) - S(q_0 + 1)$ is 0.881 ($\ell = 100$) and 0.946 ($\ell = 3200$) times the value of (18), and the ratio of Table 3 is 3.70 at $\ell = 3200$.
- *Periodic XX ring, $\ell = L/2$.* The ratio is 2.65 at $L = 24$ (our value in Table 3) and 3.63 at $L = 3200$, and $G_1 = 0.0481$ at $L = 3200$, against 0.0395 at $L = 24$.

The asymptotic amplitude is therefore approached slowly, with deviations of several per cent even at ℓ in the thousands. This puts the values at $L = 24$ in perspective. For $\Delta \neq 0$ the G_1 of Table 3 are finite-size values of the non-universal amplitudes discussed in [8]; we have not compared them with the twisted overlaps computed there.

What is reproduced and what is new. Equipartition, the $(K/\pi^2)\log \ell$ charge variance with its oscillating correction, and the $\log^{-2} \ell$ breaking are known [11, 31, 3, 27, 8]. For free fermions the amplitude of the breaking is known exactly [3], and (14) is Eq. (2.6) of [3] written with p_n. Larger sizes have been reached by DMRG and quantum Monte Carlo [16, 31, 6]. Our additions are modest. They are the integer-sampled widths B_n, for which the $1/n$ law is visible at $L \le 24$ (well for large K, less so near $\Delta = 1$); the finite-size amplitudes $G_1(\Delta)$ for $\Delta \neq 0$; the Luttinger parameter from the charge statistics, to within 0.8 % for $-0.5 \le \Delta \le 0.75$; and an exact check of the free-fermion amplitude, including how slowly it is approached.

5 Validation

All computations are in double precision. The scripts `scripts/permutational.py` and `scripts/symmetry_resolved.py` of the repository [18] compute the checks below together with the results, and the unit tests in `tests/test_permutational.py` and `tests/test_symmetry_resolved.py` repeat them at small sizes with fixed tolerances. Deviations are maximal absolute deviations unless stated otherwise.

5.1 Part A

The checks were run in the script's check mode (CPU (1 thread)).

(A1) *Exact generator.* The banded generator, with all nine terms of (5) switched on, is compared with the exact Lindblad generator on $(\mathbb{C}^2)^{\otimes N}$, compressed to $\mathcal{M}$ through Dicke bases built independently from J^2 and J_z (eigenvectors of J^2 at $m = j$, then lowering with J_-). For random permutation-invariant states it agrees to 6×10^{-16} for $N \leq 8$, and the exact generator maps $\mathcal{M}$ into itself up to round-off. An RK4 trajectory computed on $(\mathbb{C}^2)^{\otimes 7}$ gives the same blocks to 3×10^{-16}.

(A2) *Generic route.* `dynamics.lindblad_evolve` with the local jumps as `InterSectorChannel` (RK4) agrees with the banded generator to 5×10^{-17} ($N = 10$; 0.98 s against 0.18 s). The second-order Kraus integrator of the same function agrees to 7×10^{-5}.

(A3) *PIQS.* Against `qutip.piqs` [26] (QuTiP 5.3.1 [17]), whose local dephasing rate is $4\gamma_\phi$ in our convention, the generators agree to a relative 7×10^{-8}, limited by the single-precision rates of PIQS. Trajectories of (8) ($\gamma = 0.1\kappa$, $\gamma_\phi = 0.025\kappa$, $\kappa t \leq 10$) agree for $N = 10, 20, 30$ to 3×10^{-7} in m_z and to 8×10^{-9} in the final blocks. The steady states (null vectors of the two generators) agree for $N = 10, 20$ to 9×10^{-9}. For $N = 30$ one trajectory took 4.3 s here and 0.1 + 20.0 s with PIQS (Liouvillian, `mesolve`).

(A4) *Superradiance.* Pure collective decay $\mathcal{D}[J_-]$ (rate $\Gamma = 1$) from $|S, S\rangle$ is a pure death process with rates $\Gamma(S+m)(S-m+1)$, so $\int_0^\infty P(m > 0)\, dt = \sum_{m=1}^{S} [\Gamma(S+m)(S-m+1)]^{-1} = H_N/(\Gamma(N+1))$, with the harmonic number H_N. Table 4 reproduces this to 3×10^{-14} for $N \leq 200$. It also shows the superradiant delay $t_D \approx \ln N/(\Gamma N)$ and the reduction of the peak of the mean intensity below the mean-field value $\Gamma N^2/4$ by the fluctuations of the delay [7, 12].

(A5) *Product states.* Local emission, pumping and dephasing alone keep a product state a product of identical qubit states. Then $\langle J_z \rangle$, $\langle J_x \rangle$ and $\langle J^2 \rangle$ follow from the Bloch equations of one qubit, and the spin distribution is $p_j = m_j \sum_{m=-j}^{j} \lambda_+^{S+m} \lambda_-^{S-m}$ with the eigenvalues $\lambda_\pm$ of the one-qubit state. For $N = 100$ ($\gamma_-, \gamma_+, \gamma_z = 0.3, 0.1, 0.2$, initial polar angle $2\pi/3$, $t \leq 2$) the moments agree to a relative 9×10^{-14} and $\sum_j |p_j - p_j^{\text{exact}}| = 2 \times 10^{-9}$. This tests the multiplicities in (4) far beyond the reach of the full space.

(A6) *Time step.* Halving the RK4 step changes m_z by at most 1×10^{-6} and τ by at most 5×10^{-6} ($N = 30$, all ten parameter sets, $\kappa t \leq 30$).

(A7) *Unit tests* (tolerance 10^{-12} unless stated). The algebra has $\sum_j (2j+1) m_j = 2^N$; the collective operators are the restrictions of those on $(\mathbb{C}^2)^{\otimes N}$; $\Phi^*(1)$ equals $N/2 + J_z$, $N/2 - J_z$, N (also with truncation); the banded generator equals the exact one for $N = 2, \ldots, 6$; `local_jump` equals $\sum_i L_i \rho L_i^*$ compressed to $\mathcal{M}$ ($N = 5$); the generic and banded routes agree for a batch of two parameter sets; a full-space RK4 trajectory agrees to 10^{-13} ($N = 4$); the real algebra with drive about y reproduces the complex computation (10^{-13}, $N = 9$); the product-state formulas hold for $N = 30$ (10^{-10} for the moments, 10^{-8} for $\langle J^2 \rangle$ and p_j); the death-process integral holds to a relative 10^{-6} ($N = 12$); the truncated blocks satisfy $0 \leq \sigma_j^{\text{trunc}} \leq \sigma_j$ and the error bound of Section 2.2 ($N = 10$, $j_{\min} = 3$); generator and steady state agree with PIQS ($N = 6$, relative 10^{-5}); and the lifetime routine returns $\tau = 5 + \pi/4$ to within 0.06 for $m_z = -e^{-t/5} \cos 2t$, and $\tau = \infty$ for the mean-field solution without local terms.

5.2 Part B

The checks were run together with the results (full mode, CPU (2 threads)).

Table 4: Check (A4), pure collective decay from $|S,S\rangle$ ($\Gamma = 1$): the integral of $P(m>0)$ against the exact value $H_N/(N+1)$, the time t_D of the peak of the mean intensity $\Gamma\langle J_+J_-\rangle$ in units of $\ln N/N$, and the peak in units of $N^2/4$ (mean field: 1 and 1 for $N\to\infty$).

N	$\int P(m>0)\,dt$	$H_N/(N+1)$	rel. dev.	$t_D N/\ln N$	$I_{\max}/(N^2/4)$
10	0.266270	0.266270	3×10^{-14}	0.924	0.910
20	0.171321	0.171321	1×10^{-16}	1.010	0.835
50	0.088220	0.088220	2×10^{-16}	1.049	0.799
100	0.051360	0.051360	1×10^{-16}	1.055	0.789
200	0.029244	0.029244	3×10^{-16}	1.054	0.785

(B1) *Free fermions.* For $\Delta = 0$ the Jordan–Wigner transformation gives free fermions, the periodic bond carrying the sign $(-1)^{N-1}$. With the eigenvalues ν_j of the correlation matrix restricted to A [23], flux insertion [11] gives $Z_n(\alpha) = \prod_j[\nu_j^n e^{i\alpha} + (1-\nu_j)^n]$. This is a polynomial of degree ℓ in $e^{i\alpha}$, so a discrete Fourier transform on $\ell+1$ points gives $Z_n(q)$ exactly, and $\operatorname{Tr}\Pi_q\rho_A\log\rho_A$ is the transform of $\partial_n Z_n(\alpha)|_{n=1}$. For every block, $p(q)$, $\operatorname{Tr}\Pi_q\rho_A\log\rho_A$, $Z_{2,3}(q)$, $b_{1,2}$, B_1 and G_1 agree with the library to 2×10^{-14} (dense sector, $L=14$) and 2×10^{-13} (Lanczos, $L=24$), periodic and open. $S(q)$, $S_2(q)$, $S_3(q)$ agree to 4×10^{-9} where $p(q), p_n(q) > 1\times10^{-6}$ (the transform has absolute round-off). The ground energies agree with the free-fermion values to 8×10^{-13}.

(B2) *Full Hilbert space.* The dense ground state in the 2^L-dimensional space (NumPy, no symmetry used, $L=12$, $\Delta\in\{-0.5,0,0.5,1\}$, periodic and open) and the projections $\Pi_q\rho_A\Pi_q$ of the $2^\ell\times2^\ell$ matrix ρ_A give $p(q)$ and $S(q)$ for all $\ell<L$ to 7×10^{-13}. The part of ρ_A between different charges is below 2×10^{-16}.

(B3) *Three routes to ρ_A.* The pair blocks (7), `SpinChain.reduced_state` of $|\psi\rangle\langle\psi|$ and `composite.partial_trace` in the fused product give the same ρ_A to 2×10^{-16}.

(B4) *QuSpin.* QuSpin [30] (reduced density matrix projected on the charges) agrees for $L=20$, $\Delta=0.5$, periodic and open, to 5×10^{-14} in $p(q)$ and $S(q)$, and to 6×10^{-14} in the ground energy.

(B5) *Unit tests.* The Bethe-ansatz K takes the values $\frac{3}{2},1,\frac{3}{4},\frac{1}{2}$ at $\Delta=-\frac{1}{2},0,\frac{1}{2},1$. The batched Hamiltonian equals the library's sector Hamiltonian for each Δ. The free-fermion comparison (B1) holds at $L=12$ (tolerances 10^{-13} for p, Z_n; 10^{-8} for $S(q)$, $S_n(q)$). The full-space comparison (B2) holds at $L=8$ for $\Delta\in\{-0.6,0,0.3,1\}$, and the three routes (B3) agree to 10^{-14} at $L=10$. The split (13) holds to 10^{-13}, the first identity (14) to 10^{-12} and the second, with a central difference in n, to 10^{-8}. $G_1=\partial_n(nB_n)|_{n=1}$ holds to 10^{-7}, and the Gaussian form (15) is exact at $q_0\pm1$ to 10^{-12}. For the infinite XX chain at $\ell=400$, $|b_1-(\log2\ell+1+\gamma_E)/\pi^2|<10^{-6}$, $|G_1^{(b)}-1/(2\pi^2)|<2\times10^{-5}$, and $S(q_0)-S(q_0+1)$ lies between 0.9 and 1 times the value of (18); these last three are observations, not proofs.

6 Outlook

In both problems the symmetry structure turns an intractable dimension into a manageable one, 91881 entries instead of 1×10^{48} for $N=80$ spins, and a sector of dimension 2,704,156 with blocks of ρ_A of size at most $\binom{12}{6}=924$ for $L=24$. The decisive ingredients were the representation of permutation-invariant local dissipators as completely positive maps between sectors, and the sector form of a bipartite pure state.

All numbers reported here come from CPU runs: Part A in the check mode of the script ($N \leq 80$, one thread) and Part B in its full mode (two threads). The banded generator costs $O(N^3)$ per step with a number of steps that grows as N. The script has a mode with N up to 200 and longer runs designed for a GPU; GPU runs will extend N, and the batched Lanczos runs to larger L. Such runs would show whether the deviations from the additive rule (11) keep decreasing for the larger rates and keep drifting for the smallest ones. For the XXZ chain they would tell how the spread of nB_n and the amplitudes $G_1(\Delta)$ evolve with L. Other directions are local pumping and collective dephasing, which the generator (5) already contains; other definitions of the lifetime; a comparison of $G_1(\Delta)$ with the twisted overlaps of [8]; and other conserved charges, for which only the sector structure of (7) changes.

Code and data availability. The library, the scripts that compute all numbers, tables and figures, the generated data (`man/article/generated/permutational.json` and `symmetry_resolved.json`) and the tests are available at `https://github.com/andre-a-no/von_neumann_type_i` (MIT licence) [18]. The stored Part A and Part B data record revisions `c33eb35+dirty` and `b7568f6+dirty`, respectively. The uncommitted changes are not identified by these revision strings, and the computations have not been repeated after the numerical corrections listed under `Unreleased` in `CHANGELOG.md`. The results and validation figures above describe those historical runs; they do not validate the corrected implementation. See `man/REPRODUCIBILITY.md` for the reproduction status and requirements for a new numerical release.

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